Experiment 4: What the Prior Buys, and What It Costs

Insight analysis for Chapter 4, Section 4.4

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Purpose of this document. The raw write-up of Experiment 4 (`insights/experiment_4/`) reports that the pose published on a prior map tracks ground truth to 0.10–0.36 m over seven hand-driven runs, that the error is bounded rather than growing, and that two runs left the map for tens of seconds and came back. It does not say why the error is the size it is, what distinguishes the two excursions, or whether localising on the prior is better than not localising at all. This document asks those three. Every number is written by `script/exp4_nav_insights.py` into `output/compare/simulation/exp4-nav-insights/` and the tables are input from there unchanged. The script is named `exp4_nav_insights.py` because `exp4_insights.py` is the map insight script under the pre-11-November numbering, when maps were Experiment 4.

One run per cell. Seven runs, one per world-and-route combination, hand-driven and differing in shape and length. These are mechanism analyses — what kind of thing is happening — not comparisons, and the document says so where it matters.

1 Questions and method

Q1 Is the error rotational? Nothing in the slot measured heading, and heading error is what generates cross-track error downstream, so for a planner it is the primary quantity rather than a secondary one. Heading comes from the ground-truth quaternion against the `yaw_deg` column the localisation recorder samples from `map` → `base_link`, interpolated onto the same instants as the position error, with the ground-truth signal unwrapped before interpolation so that a pass through 180° does not alias. The first sample after the first fix is excluded throughout: it carries the correction still settling.

Q2 What does the prior buy? The slot's premise is that localising against a map bounds the error where dead reckoning would not. Every run already holds the unaided ORB-SLAM3 trajectory in `orb/vio.csv`, so the comparison needs no new data. The front end reports in its own arbitrary frame, so it is rigidly aligned onto ground truth first (planar Umeyama, no scale) and what remains is drift alone — the unknown initial pose given away for free. That is deliberately the kindest possible reading of unaided odometry, because the thing being tested is whether the prior adds anything beyond that free gift.

Q3 What happens at an excursion? For every unbroken stretch with the position error over 1 m lasting at least 5 s: its onset, duration and peak, and the `map` → `odom` correction before, during and after it, in both yaw and translation.

2 Insight 1: heading is accurate, and traffic is what degrades it

Median heading error is below 1.2° in all six static runs, and the 95th percentile below 3.6° in five of them. Against a chassis with a 0.6 m wheelbase driving 21–152 m aisles, that is not a limiting quantity: a degree

Table I.1: Heading error against ground truth, degrees. The first sample after the first fix is excluded: it carries the correction still settling and held both runs' apparent 160-degree maxima. "In exc." is the median while the position error is over 1 m, "out" while it is not, and r their correlation.

Run	Med.	p95	Max	Bias	Straight	Turning	In exc.	r
U static straight	0.37	0.92	1.56	+0.26	0.37	0.48	–	0.04
U static square	0.76	2.65	22.93	-1.29	0.81	0.65	1.99	0.40
U static figure-8	0.48	11.59	14.93	+2.23	0.40	0.98	11.02	0.97
I static straight	0.86	2.99	3.37	-0.93	1.12	0.69	–	0.01
I static square	1.12	3.05	3.57	-1.04	1.47	0.78	–	-0.27
I static figure-8	0.71	3.53	163.29	+4.17	0.51	1.51	–	0.61
U dynamic straight	4.85	5.26	5.78	+3.35	4.85	5.23	4.87	0.63

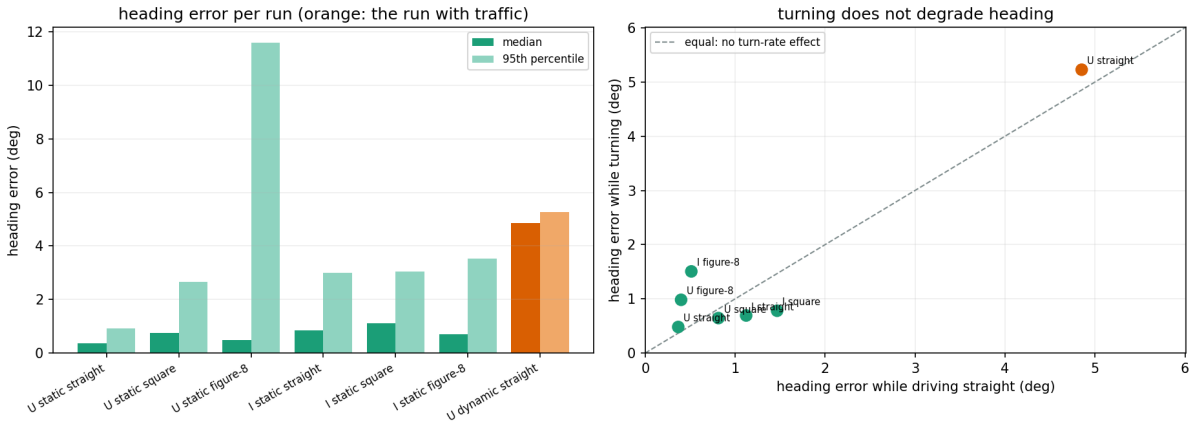


Figure I.1: Left: heading error per run, median and 95th percentile, the run with traffic in orange. Every static run sits under 1.2° median and the dynamic one at 4.85°; the gap is the finding, not the level. Right: heading error while turning against heading error while driving straight. The points sit on the diagonal, so whatever sets the floor is not the motion.

of heading error accumulates 1.7 cm of cross-track error per metre driven, and the fixes arrive far more often than that takes to matter.

The one dynamic run is the exception and it is not a small one. 4.85° median is four to thirteen times every static run, with a 3.35° bias, and it is the run whose position error is also the worst for offset and scatter. Traffic costs heading more than it costs position in relative terms — the position median is 0.298 m against a static best of 0.097 m, roughly threefold, where heading is tenfold. A moving occluder removes the distant, well-conditioned features that fix orientation while leaving the near ones that fix position, which is the shape this result has, and one dynamic run is enough to notice it and not enough to claim it.

Heading error does not rise with turn rate. The straight and turning columns differ by less than 0.8° in every run and in no consistent direction, so whatever sets the floor is not the motion.

3 Insight 2: the two excursions are two different faults

Table I.2: The two excursions, and the correction that produced each. Yaw in degrees, x in metres.

Run	Start	Dur.	Peak	Yaw before	Yaw during	x before	x during
U static square	41 s	35.5 s	6.80	-0.60	-2.10	-8.15	-13.32
U static figure-8	29 s	23.0 s	3.99	-0.50	+11.00	-8.04	-5.59

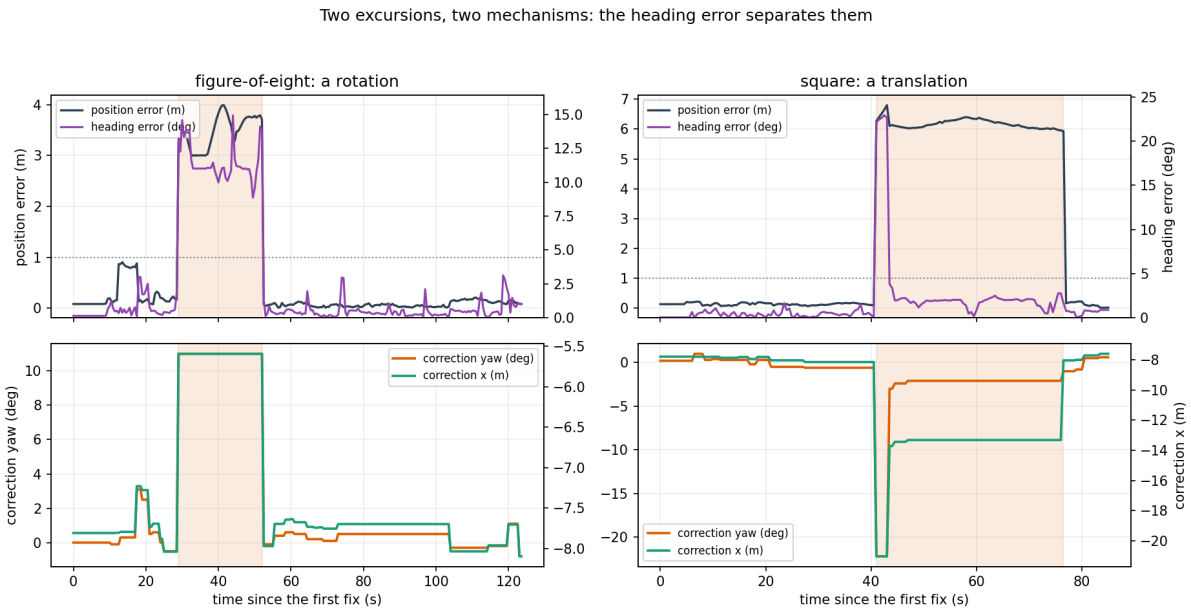


Figure I.2: The two excursions, shaded where the position error exceeds 1 m. Top row: position error with heading error on the right axis. Bottom row: the `map` → `odom` correction in yaw and in translation. Both excursions move the correction in both quantities, so the corrections alone do not separate them — the heading error does. On the figure-of-eight it rises with the position error and stays up for the whole 23 s; on the square it spikes once at the transition and falls back to about 2° while the position error holds at 6 m for 35 s. A persistent heading error is a wrong orientation; a transient one is the pose being moved.

The raw write-up reports both excursions as one phenomenon — flat shelves of error, abruptly entered and abruptly left. The corrections say otherwise.

Figure I.2 shows why the corrections alone are not the discriminator: both excursions move the correction in yaw *and* in translation. What separates them is whether the robot's heading went with it.

On the `U_static` figure-of-eight the correction's yaw moves from −0.50° to 11.00° and returns to −0.10°, its translation moving 2.45 m alongside. Heading error inside that stretch is 11.02° against 0.39°

outside it, and heading error correlates with position error at $r = 0.97$. That excursion is a rotation: the back end accepted a match that placed the robot at the right sort of place facing the wrong way, and every metre driven afterwards converted that angle into displacement until a correct fix replaced it.

On the `U_static` square the correction's yaw barely moves — −0.60° to −2.10° — while its translation shifts from −8.15 m to −13.32 m, over 5 m. Heading error inside is 1.99° against 0.42° outside, correlating at $r = 0.40$: it spikes once as the pose is moved and then falls back while the position error holds for 35 s. That excursion is a translation: the robot was placed in the wrong aisle at the right heading.

Both are perceptual aliasing and both recovered, to 0.154 m and 0.072 m. But they are the two distinct symmetries of a warehouse — the aisle pitch, and the orientation of a bay — and a defence aimed at one does not catch the other. This is the same split the November acceptance sweep found at the level of individual fixes, where the rejected matches were 73°, 143°, 23° and 24° of yaw, and it is the reason that sweep concluded geometry, proximity and a motion gate compose rather than compete.

4 Insight 3: over these distances the prior supplies a datum, not accuracy

Table I.3: The localised pose against unaided ORB-SLAM3 over the same run, metres. The front end is rigidly aligned to ground truth first, so its column is DRIFT alone with the unknown initial pose given away free – the kindest possible reading of odometry, and the one thing the prior actually supplies.

Run	Driven	Loc. med.	Loc. max	VIO med.	VIO max	VIO end
U static straight	24.6	0.162	0.24	0.045	0.17	0.03
U static square	109.2	0.188	6.80	0.176	0.35	0.18
U static figure-8	152.0	0.098	3.99	0.303	0.76	0.60
I static straight	20.9	0.097	0.61	0.302	0.47	0.23
I static square	66.8	0.361	0.69	0.283	0.72	0.39
I static figure-8	123.7	0.136	0.93	0.345	0.53	0.22
U dynamic straight	26.0	0.298	1.17	0.053	0.16	0.08

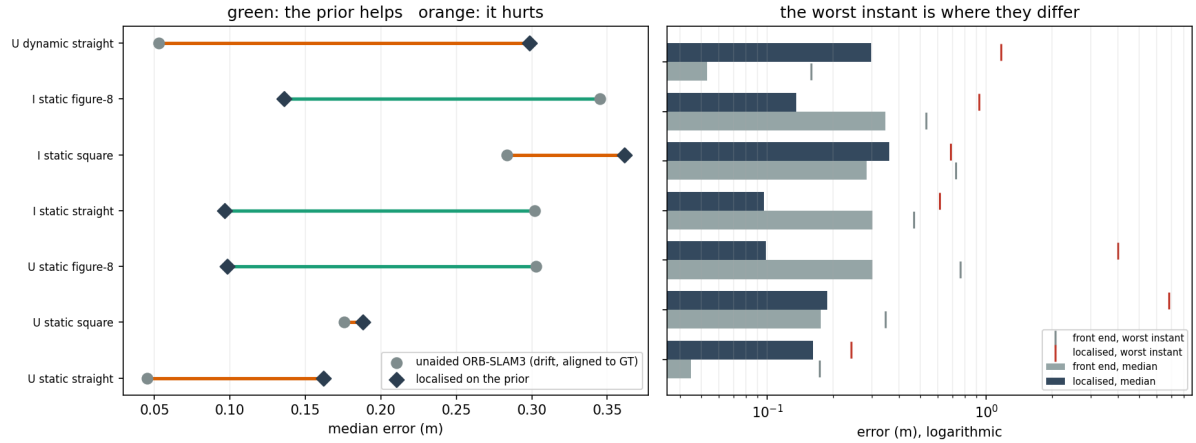


Figure I.3: Left: each run's median error, unaided front end against localised pose, joined; green where the prior helps, orange where it hurts. They split three to three with one level. Right: the same medians as bars with each column's worst instant as a tick, logarithmic. The front end never exceeds 0.76 m; the localised pose reaches 6.80 m. The asymmetry on the right, not the split on the left, is what the prior costs over these distances.

This is the uncomfortable one. Unaided ORB-SLAM3, rigidly aligned to ground truth, drifts by a median of 0.045–0.345 m over 21–152 m, never exceeds 0.76 m at any instant, and ends each run between 0.03 m and 0.60 m from truth. The localised pose, on the same runs, has a median between 0.097 m and 0.361 m and a maximum of 6.80 m.

Run by run the medians split three to three with one tie: the prior is better on both figure-of-eights and the `I_static` straight, worse on the `U_static` straight, the `I_static` square and the dynamic run, and level on the `U_static` square. There is no sense in which localising against the map improves local accuracy here. What it does do is introduce excursions an order of magnitude larger than anything the front end produced alone: 6.80 m and 3.99 m against a front-end worst of 0.76 m.

The claim that survives is narrower than the slot's premise and should replace it. The front-end column is drift *after* a rigid alignment onto ground truth, and that alignment is precisely what a deployed robot does not have. The prior supplies it: an absolute, repeatable datum in a frame that outlives the run, which is what lets a goal expressed in map coordinates mean the same thing twice, and what no amount of odometry provides. Over these route lengths that is the whole of its contribution, and it is bought at the cost of two multi-second excursions in seven runs.

The corollary is a prediction worth testing rather than a conclusion: the prior should begin to pay for itself in accuracy at the distance where front-end drift exceeds the localisation error, which these runs place somewhere beyond 152 m and the mapping routes — at 365–381 m — should already reach.

5 What these data cannot answer

Seven runs, one per cell, hand-driven on routes that differ in shape and length, so route is confounded with world and nothing here supports a comparison between layouts, between static and dynamic, or between route shapes. The dynamic condition is one run in one world. No goal was commanded in any run, so none of this speaks to planning or to the false-obstacle question that motivates the experiment. The front-end comparison of Insight 3 is one front end, ORB-SLAM3, on routes short enough that its drift stays sub-metre; it says nothing about VINS, and nothing about the distance at which the ordering reverses. Two excursions are two events, enough to show that two mechanisms exist and not enough to say which is more common.

6 Summary

Consequences for the raw write-up. Three things should change in `insights/experiment_4/`. Report heading alongside position, and report the dynamic run's heading degradation as the larger of its two costs. Replace the single "excursion" description with the two mechanisms, since they imply different defences. And restate the slot's premise: the prior's contribution over these distances is a durable frame rather than accuracy, which is a weaker claim than the one the raw write-up currently makes and is the one the data support.

Table I.4: Questions, answers and the strength of the evidence.

	Answer	Evidence	Confidence
Q1 rotation	Heading median below 1.2° in all six static runs and not a limiting quantity; the one dynamic run is 4.85°, a tenfold degradation against a threefold one in position; no turn-rate dependence	table I.1	high for static, one run for dynamic
Q2 prior's value	No accuracy gain over unaided ORB-SLAM3 at 21–152 m (3 runs better, 3 worse, 1 level) and a worst case an order of magnitude larger; what it supplies is the absolute datum the aligned front end was given free	table I.3	medium; one front end, short routes
Q3 excursions	Two events, two mechanisms: one rotational (11.0° correction, $r = 0.97$ with position error), one translational (5.2 m correction, yaw unmoved); both recovered	table I.2	medium, two events